

Saharsh Sandeep Barve

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Work Experience

Software Engineer — ML Platform

Remote, USA / Menlo Park, CA

Dragonfruit AI

Jun '25 – Present

- Built a config-defined **zero-shot CLIP classifier** framework across 5 services with prompt- and crop-embedding caches, letting customer teams stand up new attribute detectors (posture, phone use) as a JSON config rather than a model release, turning pilots and PoCs from weeks into hours while cutting inference 12.6ms → 8.5ms.
- Re-architected safety alerting from **cloud cron polling** to **event-driven edge push** over the base station's persistent WebSocket, with an at-least-once queue and cloud-side deduplication, cutting alert latency from 25s–5min to ~1–2s (>10×).
- Rebuilt on-device **multi-object tracking** for edge inference with vectorized association, dwell-based static classification and appearance re-ID, cutting per-frame cost 2.91ms → 0.77ms and scaling a single base station to 20 concurrent camera streams with zero dropped frames, while removing a recurring source of customer false alarms.
- Owned the **model development lifecycle** for 5 production detection models across annotation, training, evaluation and release, standardizing it into a gated pipeline that made shipping a model self-serve for the team and caught accuracy regressions on real deployment footage before they reached customers.
- Built a **truck turnaround monitoring** pipeline pairing YOLO detection with license-plate recognition and cloud thumbnail storage, achieving ~300ms on-prem inference per event and replacing manual gate logging for logistics customers.

Machine Learning Scientist

Bengaluru, India

Onward Assist

Jul '22 – Jul '23

- Led development of the **Nottingham scoring** models for breast-cancer histopathology grading, boosting diagnosis accuracy by 30% over the previous baseline.
- Deployed ML models to the web platform on **AWS** and **Kubernetes** with MLflow tracking and Apache Parquet data workflows, enabling reproducible and scalable serving.
- Led the team to a **top-8%** finish in the HuBMAP + HPA Kaggle competition on large-scale biomedical image analysis.

Education

University of Illinois Urbana-Champaign

Urbana-Champaign, Illinois

M.S. Computer Science, GPA: 3.96/4.0

Aug '23 – May '25

Coursework: Computer Vision, Applied Machine Learning, Large Language Models, 3D Vision, Cloud Computing, Data Management

Manipal Institute of Technology

Manipal, India

B.Tech Computer Science and Engineering, GPA: 9.25/10

Jul '18 – May '22

Coursework: Operating Systems, Parallel Programming, Computer Networks, Database Systems, Data Structures and Algorithms

Technologies

Languages: Python, C/C++, Go, TypeScript, JavaScript, SQL

ML & Vision: PyTorch, CUDA, YOLO, DETR, CLIP/MobileCLIP, OpenCV, multi-object tracking, CoreML & on-device inference

LLMs & Agents: RAG pipelines, prompt optimization, agentic workflows & tool use, MCP, VLM evaluation, LangChain

Infrastructure: Docker, Kubernetes, AWS (S3, ECS, Lambda, RDS), Celery/RabbitMQ, CI/CD

Systems: gRPC/protobuf, WebSockets, REST APIs, multi-threading, real-time & edge deployment

ML Ops & Data: Weights & Biases, MLflow, benchmark & regression design, Qdrant, FAISS, Elasticsearch, PostgreSQL

Projects

- Dynamic Prompting for LLMs** — Python, LLM APIs, GSM8K [\[github\]](#) | [report](#) Dec '24
- Built a pipeline selecting prompting strategy (zero-shot, few-shot, CoT, self-consistency CoT) per query instead of defaulting to the most expensive, reaching 89% accuracy on GSM8K with $5\times$ fewer tokens and $5\times$ faster runtime than SC-CoT.
- Visual Odometry for Autonomous Vehicles** — Python, OpenCV, ViT [\[github\]](#) | [report](#) May '24
- Benchmarked classical stereo against deep-learning visual-odometry methods on KITTI, quantifying depth-map quality and motion-tracking accuracy across approaches.
- Medical Instrument Tracking for HoloLens 2** — Computer Vision, Mixed Reality | UIUC Aug '23 – Dec '23
- Built a real-time instrument tracking system delivering mixed-reality procedural guidance to clinicians during medical procedures.
- 3D Scene Reconstruction** — NeRF, SfM, Multi-View Stereo [\[report\]](#) Dec '23
- Implemented and compared neural and classical reconstruction methods, applying Neural Kernel Surface Reconstruction to refine NeRF output and reduce noise sensitivity.

Publications

- *Social stereotypes in AI text-to-image generation* — AI and Ethics (Springer), 2026, 6(3). [\[Link\]](#)
- *Reef-Insight: A Framework for Reef Habitat Mapping with Clustering Methods Using Remote Sensing* — Information, 2023, 14, 373. [\[Link\]](#)
- *Switched Auxiliary Loss for Robust Training of Transformer Models for Histopathological Image Segmentation* — arXiv, 2023. [\[Link\]](#)

Leadership & Additional

- **Teaching Assistant, UIUC** — CS444 Deep Learning for Computer Vision, CS441 Applied Machine Learning, CS498 Cloud Computing Applications.
- **Open Source** — [BespokeLabsAI/Curator](#): synthetic data curation for post-training and data extraction.
- Recipient, J.N. Tata Endowment Scholarship; Head of Finance, IAESTE India LC Manipal (40-member organization); mentored the Onward Assist intern training program (Jan '23 cohort).